Ji 2020 cutaneous SCC — analysis summary

RUN 7 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Cell-cycle regressed out (de-novo S/G2M scores), Harmony on sample (θ=2), n_pcs=20. Embedding is sample-integrated (all 8 patients interleave; 30-NN cross-sample fraction 0.63 vs 0.78 max) and one continuous body (single radius-graph component). Without CC regression a spurious cycling axis dominates a lobe — the confound the prior linear read never removed.

Gate 2 — module geography → GO (branching)

Basal Gast/pSom hub diverges into a Gast-leaning differentiation arm (KRT1/10; Gast +0.00 vs hub −0.04) and a pSom-leaning TSK / leading-edge arm (pSom +0.06). Crest is a terminal readout enriched on the pSom (TSK) arm and depleted at the differentiation tip — uncoupled from the Gast/pSom axis (acceptable).

Gate 3 — hub cross-repression → GO (branching)

On the basal hub (n=3,756), Gast-loading and pSom-loading genes anti-correlate: mutual r = −0.38 (Puram anchor −0.49; Filbin branching −0.34). Competitive cross-repression at the branch point.

STEP FIGURES

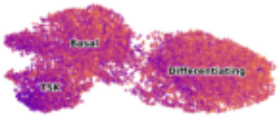
Manifold — samples



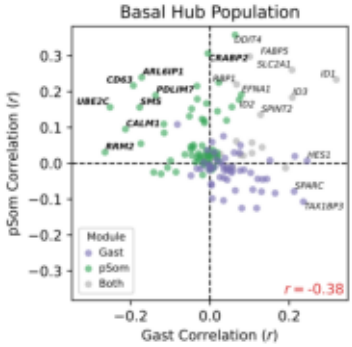
Trajectory / root



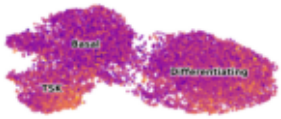
Gastrulation



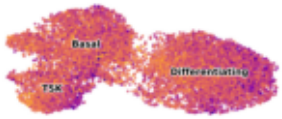
Gate 3: r = −0.38



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,523 malignant cells (8/10 patients). Geometry: BRANCHING (hub-and-spoke).

Verdict.
Pattern REPRODUCES (branching). A basal progenitor hub diverges into a keratinocyte-differentiation arm (Gast) and a TSK / leading-edge EMT arm (pSom) with hub cross-repression r = −0.38; Crest is a terminal readout on the pSom arm. Gate 1 GO / Gate 2 GO / Gate 3 GO — the third branching case in the atlas. EMT-TFs SNAI2 / PRRX2.

Parameter choices

Threshold	keep samples ≥103 (gap-finding 40→268, ~6.7×) → 8/10 patients, 10,598 → 10,531 → 10,523 (islands)	**Cell cycle**	de-novo S/G2M scores regressed out of the over-dispersed matrix (clean-embedding step)
Embedding	1,443 over-dispersed genes, PCA 50 → Harmony(sample, θ=2), n_pcs=20, UMAP spread=1.3	**Trajectory**	scFates PPT tree (Nodes=80, λ=600, σ=0.4) on X_umap; root = basal-max fork
Hub	Basal progenitor cluster (branch-point / root convergence zone); subtypes by markers + Ji calls	**EMT-TFs**	SNAI2 / PRRX2 (Gast/pSom arms)